



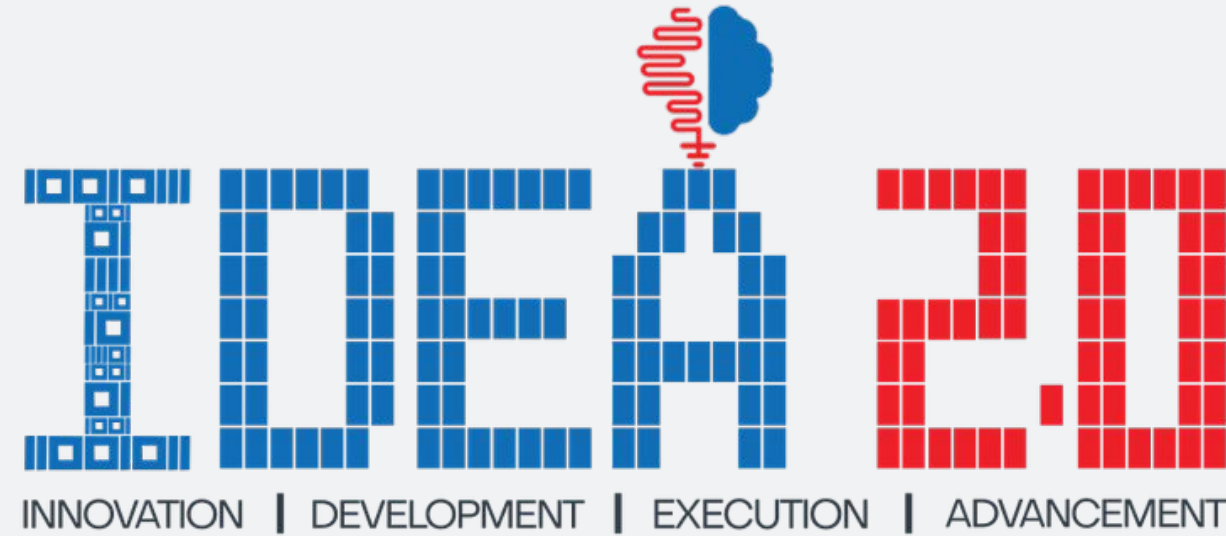
वित्तीय सेवाएं विभाग
DEPARTMENT OF
FINANCIAL SERVICES

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)



PRESENTS



A i - C S P A R C

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME (AS SUBMITTED ON PORTAL)		AgentForge	
PROBLEM STATEMENT TITLE		PS5: Unified Customer Complaint Communication Dashboard	
	NAME	GENDER	AREA OF EXPERTISE
Member 1	Prem Baba	Male	ML , Python , Agentic-AI
Member 2	Purva Bhoyar	Female	NLP , Agentic-AI
Member 3	Pranil Bankar	Male	NLP , Deep Learning
Member 4	Adhishree Shiledar	Female	Frontend Development

Proposed Solution (Describe your Idea / Solution / Prototype)

ComplaintIQ: Agentic AI-Powered Complaint Intelligence Platform

- **The Problem:**

Indian banks receive 5L+ complaints/month across 7+ fragmented channels (email, Twitter, calls, portal, branch, WhatsApp, app). Each channel is a silo — complaints get duplicated, SLA deadlines are missed, RBI penalizes the bank, and nobody connects the dots on systemic issues.

- **Our Solution:**

ComplaintIQ uses a **6-Agent AI Pipeline** that processes complaints end-to-end :

Intake → Classification → Duplicate Detection → Response Drafting → SLA Prediction → Root Cause Analysis.

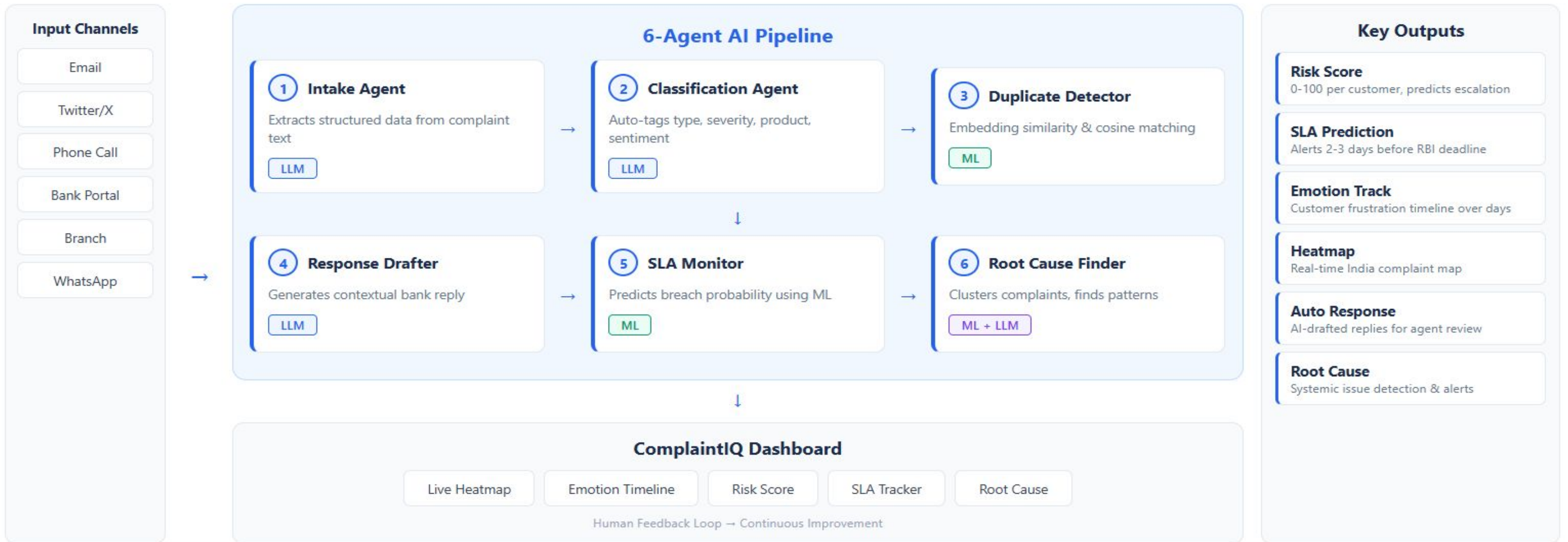
All channels feed into one unified intelligent dashboard in real-time.

- **What Makes It Unique:**

A Customer **Complaint Risk Score (0-100)** per customer that predicts escalation to RBI Ombudsman, churn risk, and negative social media impact. Plus a live **India Heatmap** showing complaint hotspots and a Customer **Emotion Timeline** tracking frustration over days.

OUTLINE OF UNIQUE & INNOVAITVE SOLUTION

ComplaintIQ — System Architecture



75-80% Custom Code

6 AI Agents

Risk Score (0-100)

Predictive SLA

Real-time Heatmap

TECHNICAL APPROACH

- Technologies to be used

Category	Technology
Language	Python 3.11
Backend	FastAPI
Frontend / Dashboard	Streamlit / React
Agent Orchestration	LangChain / CrewAI
LLM API	Anthropic Claude API
Embeddings	Sentence-Transformers
ML Models	Scikit-learn (Random Forest)
Database	PostgreSQL + ChromaDB
Visualization	Plotly, Folium (heatmap)

Methodology & Implementation Process

PHASE 1
Data

Data Layer Setup

Generate synthetic complaints using LLM + set up PostgreSQL & ChromaDB vector store

PHASE 2
Agents

Agent Development

Build 6 specialized agents individually using LangChain / CrewAI framework

PHASE 3
ML

Model Training

Train SLA prediction (Random Forest), generate embeddings (Sentence-Transformers)

PHASE 4
Pipeline

Pipeline Integration

Connect all 6 agents in sequence, test end-to-end complaint processing flow

PHASE 5
UI

Dashboard Development

Build Streamlit dashboard with Plotly charts, Folium heatmap, emotion timeline

PHASE 6
Test

Testing & Feedback Loop

Human-in-the-loop corrections, accuracy improvement, edge case handling

DEPLOY

Cloud Deployment

Deploy as web application with API endpoints, ready for demo and testing

FEASIBILITY & VIABILITY

• Analysis of the feasibility:

- All 6 agents are modular - built and tested independently
- LLM APIs are production- ready and scalable
- Sentence-Transformers and Scikit-learn are industry-standard
- No hardware dependency - runs on cloud with standard Python

• Strategies for overcoming challenges:

- Synthetic data generated using LLMs mirrors real banking complaints
- Async processing + caching solves LLM latency
- LLM multilingual capabilities handle Hindi, Marathi, English
- Human feedback loop continuously improves accuracy

• Potential challenges and risks:

Challenge	Risk Level
Real bank data is confidential	Medium - mitigated by synthetic data
LLM response latency	Low - async processing handles this
Multi-language complaints	Low - LLMs support Hindi, Marathi
Initial model accuracy	Medium - improves with feedback loop

IMPACT & BENEFITS

- Impact on target audience:

Stakeholder	Impact
For Bank	60-70% faster complaint resolution , 40% fewer SLA breaches , 500+ man-hours/month saved
For Customers	Faster resolution , consistent experience across all channels , proactive outreach before escalation
For Regulators (RBI)	Auto-generated compliance reports , 80% less manual regulatory reporting effort

- Benefits:

Type	Benefit
Economic	Saves 500+ man-hours/month, reduces RBI penalty risk from SLA breaches
Social	Multilingual support (Hindi, Marathi, English) improves banking access for non-English speakers , supports financial inclusion
Operational	Eliminates data silos across channels , Complaint Risk Score helps prioritize high-value at-risk customers

BUSINESS MODEL

- **Business Model Overview:**

- **Revenue Model:** SaaS subscription with 3 tiers:

- Basic - Classification + Response Drafting
- Professional - SLA Prediction + Heatmap
- Enterprise - Full suite + Custom Integrations

- **Target Customers:**

12 PSBs + **21** Private Banks + **40+** NBFCs + Insurance companies All RBI-mandated to have complaint systems

- **Value Proposition:**

Transforms complaint management from manual cost center into AI-powered intelligence tool

- **Commercialization & Scalability:**

Aspect	Details
Market Size	70+ Indian financial institutions, Global market \$3.2B+
Phase 1	Union Bank pilot (this hackathon)
Phase 2	Expand to PSBs via IBA network
Phase 3	Private banks and NBFCs
Partnerships	CRM providers (Salesforce, Freshworks)
Scalability	Cloud-native, each agent scales independently
Challenge	Long enterprise sales — mitigated by govt-backed pilot first

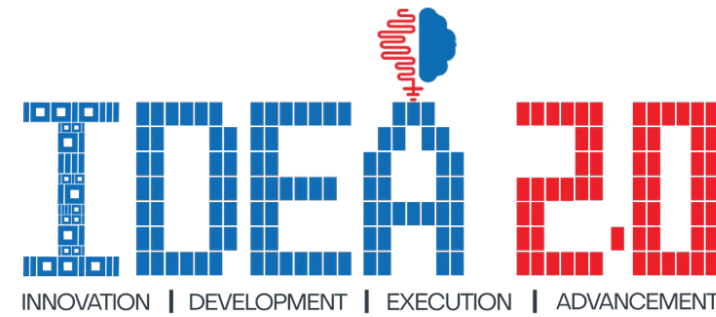


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भारत सरकार का उपक्रम



Union Bank
of India
A Government of India Undertaking



RESEARCH & REFERENCES

- **References:**

- RBI Integrated Ombudsman Scheme 2021
- RBI Master Circular on Customer Service in Banks (2024)
- LangChain Documentation — multi-agent orchestration
- Sentence-Transformers (Reimers & Gurevych, 2019)
- CrewAI Framework — agentic AI pipeline
- CFPB Consumer Complaint Database
- Anthropic Claude API Documentation

- **Mandatory 1-page summary:**

[<https://drive.google.com/drive/folders/1zSWIMTNe9RSTWaKoS4S-r8zGL-WVBjLt?usp=sharing>]